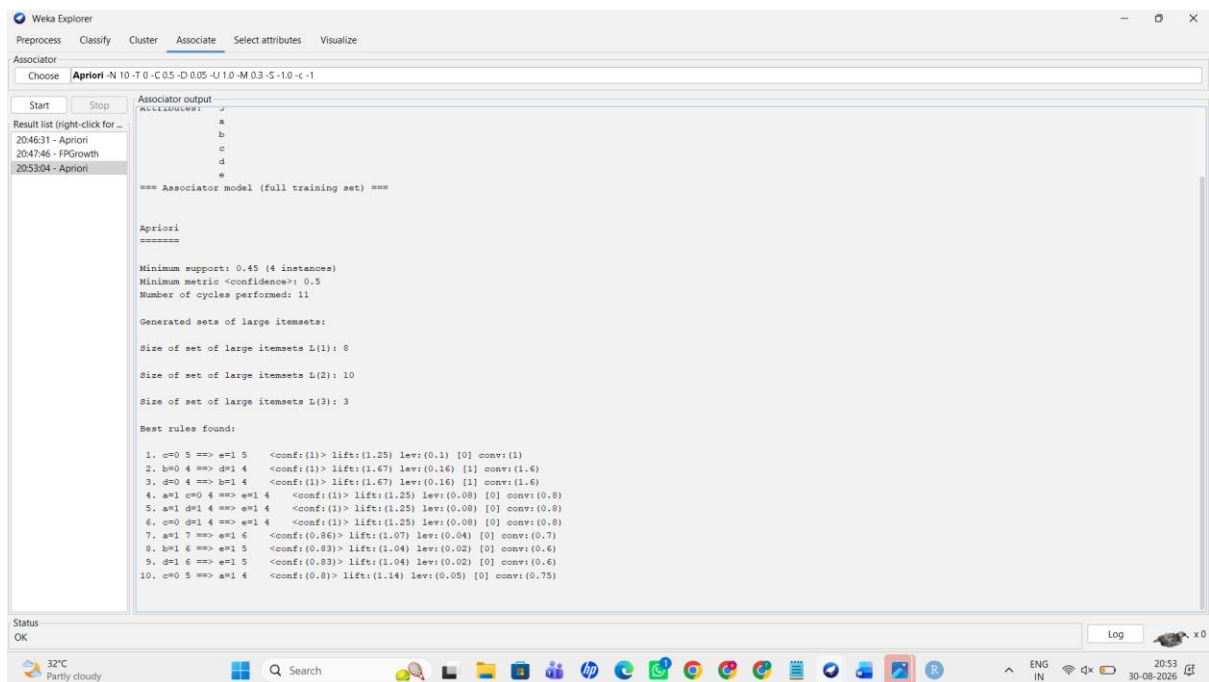
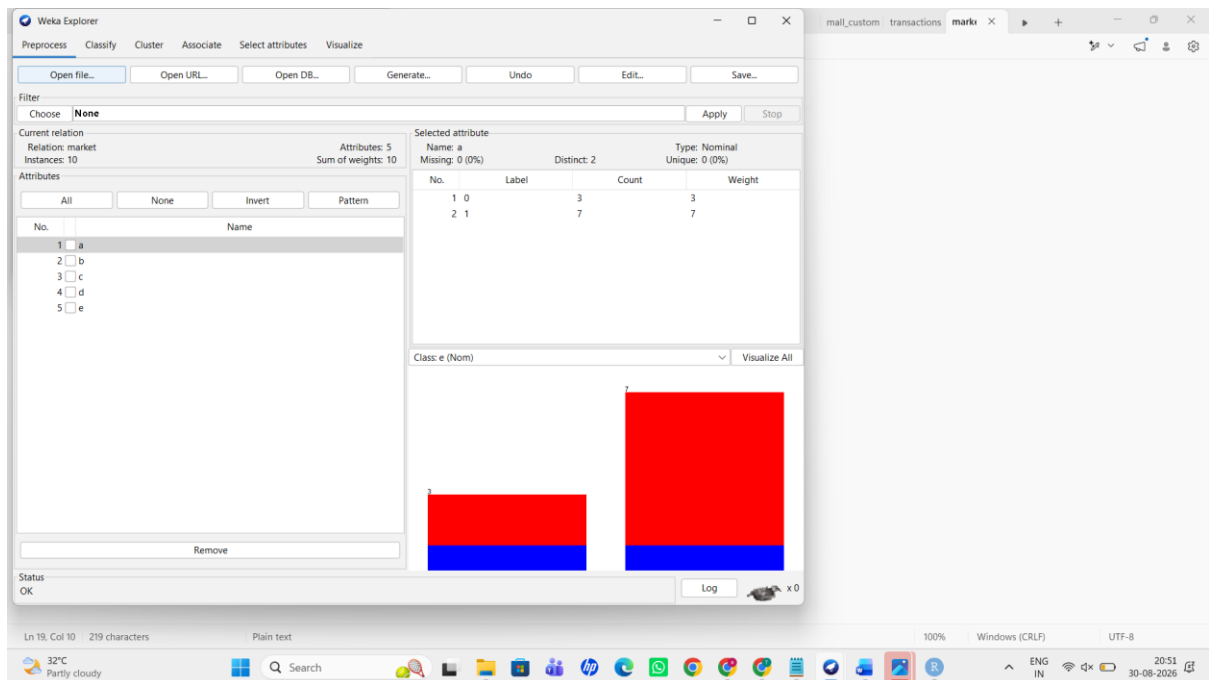


DAY 3,4 EXERCISE PROGRAM



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associate

Choose **FPGrowth** -P 2 -I 1 -N 10 -T 0 -C 0.5 -D 0.05 -U 1.0 -M 0.3

Start Stop

Result list (right-click for...)

- 204631 - Apriori
- 204746 - FPGrowth
- 205304 - Apriori
- 205401 - FPGrowth

Associator output

```

=== Run information ===

Scheme:      weka.associations.FPGrowth -P 2 -I 1 -N 10 -T 0 -C 0.5 -D 0.05 -U 1.0 -M 0.3
Relation:     market
Instances:    10
Attributes:   5
a
b
c
d
e

=== Associator model (full training set) ===

FPGrowth found 16 rules (displaying top 10)

1. [a=1, d=1]: 4 ==> [a=1]: 4 <conf:(1)> lift:(1.25) lev:(0.08) conv:(0.8)
2. [a=1]: 7 ==> [a=1]: 6 <conf:(0.86)> lift:(1.07) lev:(0.04) conv:(0.7)
3. [d=1]: 6 ==> [a=1]: 5 <conf:(0.83)> lift:(1.04) lev:(0.02) conv:(0.6)
4. [b=1]: 6 ==> [a=1]: 5 <conf:(0.83)> lift:(1.04) lev:(0.02) conv:(0.6)
5. [a=1, d=1]: 5 ==> [a=1]: 4 <conf:(0.8)> lift:(1.14) lev:(0.05) conv:(0.75)
6. [a=1]: 8 ==> [a=1]: 6 <conf:(0.75)> lift:(1.07) lev:(0.04) conv:(0.8)
7. [d=1]: 6 ==> [a=1]: 4 <conf:(0.67)> lift:(0.95) lev:(-0.02) conv:(0.6)
8. [b=1]: 6 ==> [a=1]: 4 <conf:(0.67)> lift:(0.95) lev:(-0.02) conv:(0.6)
9. [a=1, a=1]: 6 ==> [d=1]: 4 <conf:(0.67)> lift:(1.11) lev:(0.04) conv:(0.8)
10. [d=1]: 6 ==> [a=1, a=1]: 4 <conf:(0.67)> lift:(1.11) lev:(0.04) conv:(0.8)

```

Status OK

Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter Choose **None** Apply Stop

Current relation: buys_computer
Instances: 14

Attributes: 5 Sum of weights: 14

Attributes

All None Invert Pattern

No.	Name
1	age
2	income
3	student
4	credit_rating
5	buys_computer

Remove

Selected attribute

Name: age
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 0 (0%)

No.	Label	Count	Weight
1	<=30	5	5
2	31_40	4	4
3	>40	5	5

Class: buys_computer (Nom)

Visualize All

Status OK

Log

Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **J48 - C 0.25 - M 2**

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: **10**
☐ Percentage split % **66**
More options...

(Nom) buys_computer

Start Stop

Result list (right-click for options):
20:58:30 - trees.J48

Classifier output:
age = 31...40: yes (9.0)
age = 340
| credit_rating = fair: yes (3.0)
| credit_rating = excellent: no (2.0)
Number of Leaves : 5
Size of the tree : 8
Time taken to build model: 0.02 seconds

=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 7 50 %
Incorrectly Classified Instances 7 50 %
Kappa statistic -0.0426
Mean absolute error 0.4367
Root mean squared error 0.5904
Relative absolute error 87.5 %
Root relative squared error 121.2987 %
Total Number of Instances 14

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	ROC Area	PRC Area	Class
Weighted Avg.	0.556	0.600	0.625	0.556	0.588	-0.043	0.633	yes
	0.400	0.444	0.333	0.400	0.364	-0.043	0.633	no

=== Confusion Matrix ===
a b <-- classified as
5 4 | a = yes
3 2 | b = no

Status: OK

32°C Partly cloudy

Log

Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes**

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: **10**
☐ Percentage split % **66**
More options...

(Nom) buys_computer

Start Stop

Result list (right-click for options):
20:58:30 - trees.J48
20:58:12 - bayes.NaiveBayes

Classifier output:
age = 31...40: yes (9.0)
age = 340
| credit_rating = fair: yes (3.0)
| credit_rating = excellent: no (2.0)
Number of Leaves : 5
Size of the tree : 8
Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 8 57.1429 %
Incorrectly Classified Instances 6 42.8571 %
Kappa statistic -0.0244
Mean absolute error 0.4374
Root mean squared error 0.4916
Relative absolute error 91.8631 %
Root relative squared error 99.6492 %
Total Number of Instances 14

=== Detailed Accuracy By Class ===

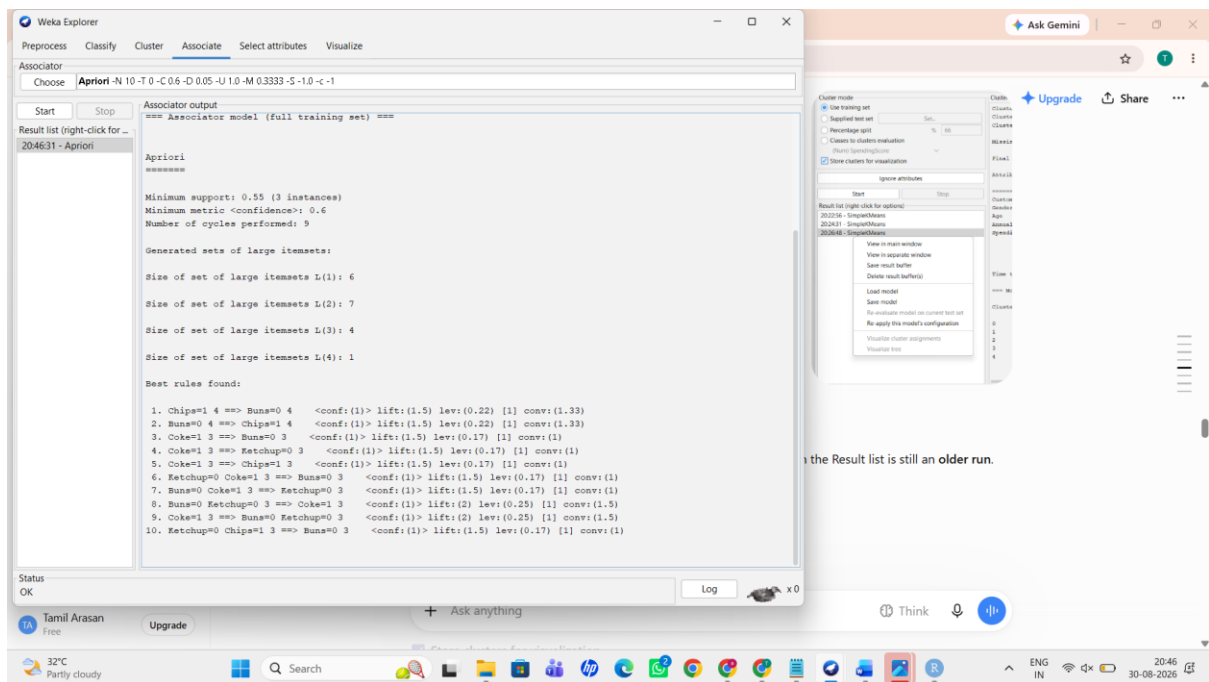
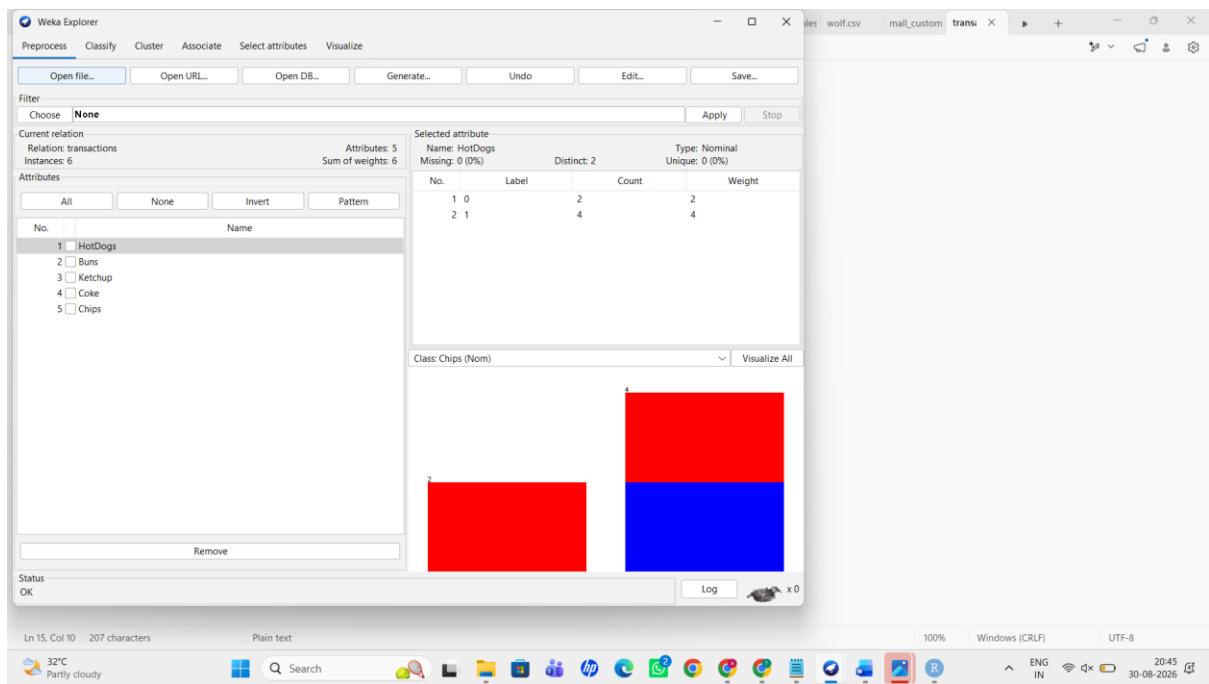
	TP Rate	FP Rate	Precision	Recall	F-Measure	ROC Area	PRC Area	Class
Weighted Avg.	0.778	0.800	0.636	0.778	0.700	-0.026	0.578	yes
	0.200	0.222	0.333	0.200	0.250	-0.026	0.578	no

=== Confusion Matrix ===
a b <-- classified as
7 2 | a = yes
4 1 | b = no

Status: OK

32°C Partly cloudy

Log



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose **FPGrowth** -P 2 -I -1 -N 10 -T 0 -C 0.6 -D 0.05 -U 1.0 -M 0.3333

Start Stop

Result list (right-click for ...)

204631 - Apriori

204746 - FPGrowth

Associator output

```
=== Run information ===
Scheme: weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.6 -D 0.05 -U 1.0 -M 0.3333
Relation: transactions
Instances: 6
Attributes: 5
HotDogs
Buns
Ketchup
Coke
Chips

=== Associator model (full training set) ===
FPGrowth found 8 rules (displaying top 8)
1. [Buns=1]: 2 ==> [HotDogs=1]: 2 <conf:(1)> lift:(1.5) lev:(0.11) conv:(0.67)
2. [Coke=1]: 3 ==> [Chips=1]: 3 <conf:(1)> lift:(1.5) lev:(0.17) conv:(1)
3. [HotDogs=1, Chips=1]: 2 ==> [Coke=1]: 2 <conf:(1)> lift:(1.5) lev:(0.17) conv:(1)
4. [HotDogs=1, Coke=1]: 2 ==> [Chips=1]: 2 <conf:(1)> lift:(1.5) lev:(0.11) conv:(0.67)
5. [Chips=1]: 4 ==> [Coke=1]: 3 <conf:(0.75)> lift:(1.5) lev:(0.17) conv:(1)
6. [Coke=1]: 3 ==> [HotDogs=1]: 2 <conf:(0.67)> lift:(1) lev:(0) conv:(0.5)
7. [Coke=1]: 3 ==> [HotDogs=1, Chips=1]: 2 <conf:(0.67)> lift:(2) lev:(0.17) conv:(1)
8. [Chips=1, Coke=1]: 3 ==> [HotDogs=1]: 2 <conf:(0.67)> lift:(1) lev:(0) conv:(0.5)
```

Status OK

Ask anything

Ask Gemini

Upgrade Share

Think

32°C Partly cloudy

Search

ENG IN

20:47 30-08-2026

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter

Choose **None** Apply Stop

Current relation

Relation: mall_customers

Instances: 5

Attributes: 5

Sum of weights: 5

Attributes

All None Invert Pattern

No. Name

1 ☐ CustomerID

2 ☐ Gender

3 ☐ Age

4 ☒ AnnualIncome

5 ☒ SpendingScore

Remove

Selected attribute

Name: SpendingScore

Missing: 0 (0%)

Distinct: 5

Type: Numeric

Unique: 5 (100%)

Statistic	Value
Minimum	6
Maximum	81
Mean	48.6
StdDev	30.973

Class: SpendingScore (Num) Visualize All

5

6 43.5 81

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -O -num-s

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation (Num) SpendingScore

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

20:22:56 - SimpleKMeans

20:24:31 - SimpleKMeans

20:26:48 - SimpleKMeans

20:29:32 - SimpleKMeans

Cluster output

Cluster 2: 1, Male, 19, 15, 39

Cluster 3: 3, Female, 20, 16, 6

Cluster 4: 5, Female, 31, 17, 40

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster#	0	1	2	3	4
	(5.0)		(1.0)	(1.0)	(1.0)	(1.0)	(1.0)
CustomerID	3	4	2	1	3	5	
Gender	Female	Female	Male	Male	Female	Female	
Age	22.8	23	21	19	20	31	
AnnualIncome	15.8	16	15	15	16	17	
SpendingScore	48.6	77	81	39	6	40	

Time taken to build model (full training data) : 0 seconds

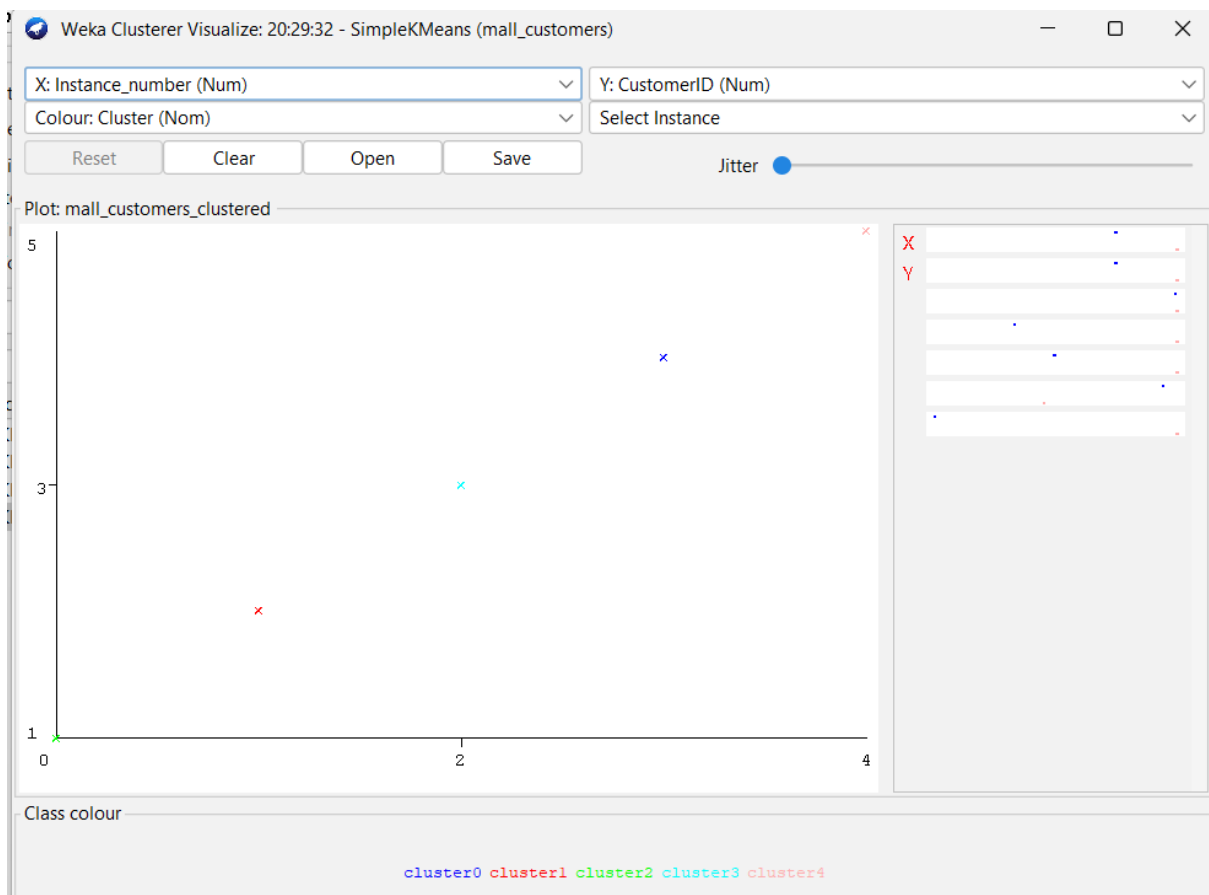
=== Model and evaluation on training set ===

Clustered Instances

```

0      1 ( 20%)
1      1 ( 20%)
2      1 ( 20%)
3      1 ( 20%)
4      1 ( 20%)

```



2.

Clusterer

Choose **EM** -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split %

☐ Classes to clusters evaluation

(Num) Unique_Shows_Per_Month ▾

☒ Store clusters for visualization

Ignore attributes

Start

Result list (right-click for options)

20:57:57 - EM

Clusterer output

Attribute	Cluster	
	0 (0.57)	1 (0.43)
=====		
Minutes_Watched		
mean	148.9823	45.9288
std. dev.	42.5173	15.6697
Sessions_Per_Week		
mean	10.4937	3.8196
std. dev.	2.5811	1.2243
Unique_Shows_Per_Month		
mean	10.3706	2.8196
std. dev.	3.2752	1.2243

Time taken to build model (full training data) : 0.04 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	9 (56%)
1	7 (44%)

Log likelihood: -9.62625

3

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **HierarchicalClusterer** -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance" -R first-last

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split %

☐ Classes to clusters evaluation

(Num) Unique_Shows_Per_Month ▾

☒ Store clusters for visualization

Ignore attributes

Start

Result list (right-click for options)

20:57:57 - EM

20:58:47 - HierarchicalClusterer

Clusterer output

=== Run information ===

Scheme: weka.clusterers.HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance" -R first-last

Relation: streaming

Instances: 16

Attributes: 3

Minutes_Watched

Sessions_Per_Week

Unique_Shows_Per_Month

Test mode: evaluate on training data

=== Clustering model (full training set) ===

Cluster 0

(((((2.0:0.05128,2.0:0.05128):0.05369,(3.0:0.02564,3.0:0.02564):0.07933):0,1.0:0.1045):0.07933):0.07933)

Cluster 1

(15.0:0.1445,16.0:0.1445)

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	14 (88%)
1	2 (13%)

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **FPGrowth** -P 2 -I -1 -N 10 -T 0 -C 0.75 -D 0.05 -U 1.0 -M 0.5

Start Stop

Result list (right-click for ...)

- 20:50:16 - Apriori
- 20:52:12 - FPGrowth

Associator output

```

=== Run information ===

Scheme:      weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1
Relation:    market_transactions
Instances:   6
Attributes:  5
              Hot_Dogs
              Buns
              Ketchup
              Coke
              Chips

=== Associator model (full training set) ===

FPGrowth found 6 rules (displaying top 6)

1. [Buns=1]: 2 ==> [Hot_Dogs=1]: 2 <conf:(1)> lift:(1.5) lev:(0.11) conv:(0.67)
2. [Coke=1]: 3 ==> [Chips=1]: 3 <conf:(1)> lift:(1.5) lev:(0.17) conv:(1)
3. [Hot_Dogs=1, Chips=1]: 2 ==> [Coke=1]: 2 <conf:(1)> lift:(2) lev:(0.17) conv:(1)
4. [Hot_Dogs=1, Coke=1]: 2 ==> [Chips=1]: 2 <conf:(1)> lift:(1.5) lev:(0.11) conv:(0.67)
5. [Hot_Dogs=1, Ketchup=1]: 1 ==> [Buns=1]: 1 <conf:(1)> lift:(3) lev:(0.11) conv:(0.67)
6. [Ketchup=1, Buns=1]: 1 ==> [Hot_Dogs=1]: 1 <conf:(1)> lift:(1.5) lev:(0.06) conv:(0.33)
  
```

Status OK Log x 0

reprocess Classify Cluster Associate **Select attributes** Visualize

Open file... Open URL... Open DB... Generate... Undo

Iter

Choose **None**

Current relation

Relation: Yogurt
Instances: 5

Attributes

Attributes: 6
Sum of weights: 5

Selected attribute

Name: Bread
Missing: 0 (0%)
Distinct: 2

Statistic

Minimum
Maximum
Mean
StdDev

Class: Yogurt (Num)

5

No.	Name
1	☐ Bread
2	☐ Cheese
3	☐ Egg
4	☐ Juice
5	☐ Milk
6	☐ Yogurt

Associator

Choose **FPGrowth** -P 2 -I -1 -N 10 -T 0 -C 0.75 -D 0.05 -U 1.0 -M 0.5

Start Stop

Result list (right-click for options)

Associator output

Generated sets of large itemsets:

Size of set of large itemsets L(1): 6

Size of set of large itemsets L(2): 7

Size of set of large itemsets L(3): 4

Size of set of large itemsets L(4): 1

Best rules found:

1. Chips=1 4 ==> Buns=0 4 <conf:(1)> lift:(1.5) lev:(0.22) [1] conv:(1.33)

2. Buns=0 4 ==> Chips=1 4 <conf:(1)> lift:(1.5) lev:(0.22) [1] conv:(1.33)

3. Coke=1 3 ==> Buns=0 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

4. Coke=1 3 ==> Ketchup=0 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

5. Coke=1 3 ==> Chips=1 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

6. Ketchup=0 Coke=1 3 ==> Buns=0 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

7. Buns=0 Coke=1 3 ==> Ketchup=0 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

8. Buns=0 Ketchup=0 3 ==> Coke=1 3 <conf:(1)> lift:(2) lev:(0.25) [1] conv:(1.5)

9. Coke=1 3 ==> Buns=0 Ketchup=0 3 <conf:(1)> lift:(2) lev:(0.25) [1] conv:(1.5)

10. Ketchup=0 Chips=1 3 ==> Buns=0 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

11. Buns=0 Ketchup=0 3 ==> Chips=1 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

12. Coke=1 Chips=1 3 ==> Buns=0 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

13. Buns=0 Coke=1 3 ==> Chips=1 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

14. Coke=1 3 ==> Buns=0 Chips=1 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

15. Coke=1 Chips=1 3 ==> Ketchup=0 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

16. Ketchup=0 Chips=1 3 ==> Coke=1 3 <conf:(1)> lift:(2) lev:(0.25) [1] conv:(1.5)

17. Ketchup=0 Coke=1 3 ==> Chips=1 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

18. Coke=1 3 ==> Ketchup=0 Chips=1 3 <conf:(1)> lift:(2) lev:(0.25) [1] conv:(1.5)

19. Ketchup=0 Coke=1 Chips=1 3 ==> Buns=0 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

20. Buns=0 Coke=1 Chips=1 3 ==> Ketchup=0 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

Classifier

Choose **J48** -C 0.25 -M 2

Test options

☒ Use training set

☐ Supplied test set Set...

☐ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Gender

Start Stop

Result list (right-click for options)

21:38:38 - trees.J48

Classifier output

Size of the tree : 3

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances 9 90 %

Incorrectly Classified Instances 1 10 %

Kappa statistic 0.8

Mean absolute error 0.1667

Root mean squared error 0.2887

Relative absolute error 33.3333 %

Root relative squared error 57.735 %

Total Number of Instances 10

=== Detailed Accuracy By Class ===

TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class

1.000 0.200 0.833 1.000 0.909 0.816 0.900 0.833 Male

0.800 0.000 1.000 0.800 0.889 0.816 0.900 0.900 Female

Weighted Avg. 0.900 0.100 0.917 0.900 0.899 0.816 0.900 0.867

=== Confusion Matrix ===

a b <-- classified as

5 0 | a = Male

1 4 | b = Female

ChooseJRip -F 3 -N 2.0 -O 2 -S 1

Test options

☒ Use training set

☐ Supplied test set

Set...

☐ Cross-validation

Folds10

☐ Percentage split

%66

More options...

(Nom) Gender

StartStop

Result list (right-click for options)

21:38:38 - trees.J48

21:40:19 - rules.JRip

Classifier output

Number of Rules : 2

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances

990%

Incorrectly Classified Instances

110%

Kappa statistic

0.8

Mean absolute error

0.1667

Root mean squared error

0.2887

Relative absolute error

33.3333%

Root relative squared error

57.735%

Total Number of Instances

10

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.200	0.833	1.000	0.909	0.816	0.900	0.833	Male
	0.800	0.000	1.000	0.800	0.889	0.816	0.900	0.900	Female
Weighted Avg.	0.900	0.100	0.917	0.900	0.899	0.816	0.900	0.867	

=== Confusion Matrix ===

a b <-- classified as

5 0 | a = Male

1 4 | b = Female

Classifier

ChooseJRip -F 3 -N 2.0 -O 2 -S 1

Test options

☒ Use training set

☐ Supplied test set

Set...

☐ Cross-validation

Folds10

☐ Percentage split

%66

More options...

(Nom) Gender

StartStop

Result list (right-click for options)

21:38:38 - trees.J48

21:40:19 - rules.JRip

Classifier output

Number of Rules : 2

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances

990%

Incorrectly Classified Instances

110%

Kappa statistic

0.8

Mean absolute error

0.1667

Root mean squared error

0.2887

Relative absolute error

33.3333%

Root relative squared error

57.735%

Total Number of Instances

10

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.200	0.833	1.000	0.909	0.816	0.900	0.833	Male
	0.800	0.000	1.000	0.800	0.889	0.816	0.900	0.900	Female
Weighted Avg.	0.900	0.100	0.917	0.900	0.899	0.816	0.900	0.867	

=== Confusion Matrix ===

a b <-- classified as

5 0 | a = Male

1 4 | b = Female

Select the attribute to use as the class

Classifier

Choose **JRip -F 3 -N 2.0 -O 2 -S 1**

Test options

☒ Use training set
 ☐ Supplied test set
☐ Cross-validation Folds
☐ Percentage split %

(Nom) Gender

Result list (right-click for options)

21:38:38 - trees.J48

21:40:19 - rules.JRip

Classifier output

Number of Rules : 2

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances	9	90
Incorrectly Classified Instances	1	10
Kappa statistic	0.8	
Mean absolute error	0.1667	
Root mean squared error	0.2887	
Relative absolute error	33.3333 %	
Root relative squared error	57.735 %	
Total Number of Instances	10	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Me
--	---------	---------	-----------	--------	------

Classifier

Choose **JRip -F 3 -N 2.0 -O 2 -S 1**

Test options

☒ Use training set
 ☐ Supplied test set
☐ Cross-validation Folds
☐ Percentage split %

(Nom) Gender

Result list (right-click for options)

21:38:38 - trees.J48

21:40:19 - rules.JRip

Classifier output

Number of Rules : 2

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances	9	90	%
Incorrectly Classified Instances	1	10	%
Kappa statistic	0.8		
Mean absolute error	0.1667		
Root mean squared error	0.2887		
Relative absolute error	33.3333 %		
Root relative squared error	57.735 %		
Total Number of Instances	10		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.200	0.833	1.000	0.909	0.816	0.900	0.833	Male
	0.800	0.000	1.000	0.800	0.889	0.816	0.900	0.900	Female
Weighted Avg.	0.900	0.100	0.917	0.900	0.899	0.816	0.900	0.867	

=== Confusion Matrix ===

```

a b  <-- classified as
5 0 | a = Male
1 4 | b = Female

```

```
> x <- c(100,70,60,90,90)
> print((x-min(x))/(max(x)-min(x)))
[1] 1.00 0.25 0.00 0.75 0.75
> print((x-mean(x))/sd(x))
[1] 1.0954451 -0.7302967 -1.3388774 0.4868645 0.4868645
> mad <- mean(abs(x-mean(x)))
> print((x-mean(x))/mad)
[1] 1.3235294 -0.8823529 -1.6176471 0.5882353 0.5882353
> print(x/1000)
[1] 0.10 0.07 0.06 0.09 0.09

> AvgSpeed <- c(/8,81,82,/4,83,82,/,80,/0)
> TotalTime <- c(39,37,36,42,35,36,40,38,46)
> sd(AvgSpeed)
[1] 4.304391
> sd(TotalTime)
[1] 3.492054
> var(AvgSpeed)
[1] 18.52778
> var(TotalTime)
```